ANSWER PRIVATE ACADEMY

2019/2020MALAWI SCOOL CERTIFICATE OF EDUCATION MOCK EXAMINATIONS

CHEMISTRY

September, 2020 FORM IV **Subject No: M022/II**

PAPER II

(**40** Marks)

Practical

1. This paper contains **7** pages. **Please check**.
2. Answer **all** questions in the blank spaces provided on the question paper. The maximum number of marks for each answer is indicated against each question. A pencil should be used for all drawings.
3. Write your **examination number** on each page of this paper.
4. In the table provided on this page, **tick** against the question number you have answered.
5. Hand in your examination paper to the invigilator when time is called to stop writing.

Question number	Tick question 1 – 4 if answered	Do not write in these columns	
1			
2			
3			
4			

SECTION A (20Marks)

1. Construct a flow diagram that could be used to identify ethanol, propanone, acetic acid ethanol and hexane using tests that make use of distilled water, sodium hydroxide solution, phenolphthalein indicator, Brady's solution and Tollens reagent.

2. i. In a laboratory, labels fell off from bottles containing solutions of zinc ions, copper (ii) ions, Iron(ii) ions and irons (iii) ions, labelled P,Q, R and S but not in that order. A Lab technician at answer private academy conducted an experiment using aqueous ammonia solution to identify the contents of the bottles. Below are the results of her experiment.

TABLE OF RESULTS

Solution	3 drops of Ammonia	Excess Ammonia
P	Brue precipitate	Dissolves
Q	White precipitate	Dissolves
R	Green Precipate	Insoluble
S	Red brown Precipitate	insoluble

Identify the ions

P ____

Q ____

R ____

S ____(4 marks)

ii. In the experiment give any **two** variables that have been kept constant.

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____(2 marks)

b. Explain how the following chemicals can be used to test for the presence of water in the substance.

i. Anhydrous cobalt (ii) chloride paper

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____(2 marks)

ii. Anhydrous copper (II) sulphate.

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____(2 marks)

SCTION B (20 marks)

3. You are provided with 100ml conical flasks, white paper with a black cross(X), stirring rod, measuring cylinder, kitchen balance, stopwatch, 2M hydrochloric acid, and sodium thiosulphate ($Na_2S_2O_3$).

a. Pour 100ml of distilled water into each of the comical flasks labeled A, B and C
b. Put 1g of sodium thoiosuphate into flask A, 2g into flasks B and 3g into flask C.
c. Stir the mixture of each flask using the stering rod to dissolve
d. Place flask A on a white paper with a black cross (X) on it.
e. Add 5ml of 2M hydrochloric acid into the flask and start the stopwatch. Swirl the contents of the flask.
f. Record time taken for the cross to be obscured (When you can not see it anymore) in the table of results below.
g. Repeat steps d – g using 2g(flask B) and 3g (flask C) of sodium thiosulphate.

TABLE OF RESULTS

Amount of sodium thiosulphate (g)	**Time taken for the cross to disappear (in seconds)**
1	
2	
3	

(3marks)

h. Plot a graph of amount of sodium thiosulphate used against time taken for the cross to disappear.

(4 marks)

i. Write a balanced chemical equation between sodium thiosulphate and hydrochloric acid

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____(3 marks)

4. You are provided with four beakers, distilled water, a measuring cylinder, sand paper and solutions of copper sulphate, iron sulphate, magnesium sulphate and aluminum nitrate. You are also provided with pieces of copper, Iron, magnesium and aluminium metals

a. Pour about 2ml of copper sulphate solution into each of the four beakers.
b. Clean the copper, Iron, magnesium and aluminum metals using sand paper.
c. Put a piece of each metal into each of the four beakers containing copper sulphate solution.
d. Observe the contents of the beakers for 2 to 3 minutes
e. record the results in the table of results below by indicating "Reaction" or "no reaction"
f. Rinse the beakers with distilled water
g. Repeat steps a – f using solutions of iron sulphate, magnesium sulphate and aluminum nitrate.

Table 2

Metals / **Solutions**	**copper**	**iron**	**magnesium**	**Aluminium**
Copper Sulphate				
Iron Sulphate				
Magnesium Sulphate				
Aluminium nitrate				

(6marks)

h. Use the results to arrange the metals in order of increasing reactivity.

(4marks)

END OF QUESTION PAPER